STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-AXIOMS-AB

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2570-2579 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 2581-2600, the complete author proof (18 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 31 of queue_sdga.json (pb=866, stmt 2570-2579, proof 2581-2600).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 09P4: sources/stacks-sdga/pinned-original-source.tex, source lines 2600–2600. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09P5: sources/stacks-sdga/pinned-original-source.tex, source lines 2600–2600. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FRY: sources/stacks-sdga/pinned-original-source.tex, source lines 2363–2379. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\mathcal{A}$ be a sheaf of differential graded algebras on $(\mathcal{C}, \mathcal{O})$. The differential graded category $Mod^{dg}(\mathcal{A}, d)$ satisfies axioms (A) and (B) of Differential Graded Algebra, Section 09P5.*

Proof. Suppose given differential graded $\mathcal{A}$-modules $\mathcal{M}$ and $\mathcal{N}$. Consider the differential graded $\mathcal{A}$-module $\mathcal{M} \oplus \mathcal{N}$ defined in the obvious manner. Then the coprojections $i : \mathcal{M} \to \mathcal{M} \oplus \mathcal{N}$ and $j : \mathcal{N} \to \mathcal{M} \oplus \mathcal{N}$ and the projections $p : \mathcal{M} \oplus \mathcal{N} \to \mathcal{N}$ and $q : \mathcal{M} \oplus \mathcal{N} \to \mathcal{M}$ are morphisms of differential graded $\mathcal{A}$-modules. Hence i, j, p, q are homogeneous of degree 0 and closed, i.e., $\mathrm{d}(i) = 0$, etc. Thus this direct sum is a differential graded sum in the sense of Differential Graded Algebra, Definition 09P4. This proves axiom (A).

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Axiom (B) was shown in Section 0FRY. □